

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] FEBRUARY-2023

Paper Code: AIDS/ AIML/IOT-201

Subject: Data Structures

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q.No1 which is compulsory.
Select one question from each unit. Scientific calculators are allowed.

Q1. Attempt the following questions:

(2.5x6 = 15)

- What is the condition to check overflow and underflow condition in a queue?
- List the advantages of linked lists over arrays.
- Evaluate the following postfix expression using stacks: 320, 10, *, 10, 60, 100, *, /
- The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?
- Discuss the time complexity to find the n th node from a linked list.
- Find the root of each of the following binary trees:
 - Tree with post order traversal: ACBDF
 - Tree with preorder traversal: PBCDFEQ

Unit-I

Q2.

- Convert the given expression into prefix and postfix expression using stack:
 $(P+Q*R)-(S+T)/U$. (7)
- Given an array of 8 elements [2,4,5,6,7,8,3,9]. Write the program to insert element '1' at the beginning of the array. (8)

OR

Q3.

- Write a program to delete an element from Circular Queue. (5)
- Consider the following operation performed on a stack of size 5. (5)

```
Push(1);  
Pop();  
Push(2);  
Push(3);  
Pop();  
Push(4);  
Pop();  
Pop();  
Push(5);
```

After the completion of all operation, find the number of elements present

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P1/-

P.T.O.

[-2-]

- c) Differentiate between stacks and queues with suitable examples. (5)

Unit-II

Q4.

- a) Write an algorithm to implement binary search on given data: 11, 22, 30, 33, 40, 44, 55, 60, 66, 77, 80, 88, 99 for the item 40. Discuss the complexity of binary search. (7)
- b) Show step by step procedure of quick sort on $A = \{9, -3, 5, 2, 6, 8, -6, 1, 3\}$. (8)

OR

Q5.

- a) Demonstrate the step by step process of insertion sort of the given array: {12, 11, 13, 5, 6} (7)
- b) Define a circular linked list. Write an algorithm to delete a node from the beginning of the circular linked list. (8)

Unit-III

Q6.

- a) Construct the AVL tree if following elements are inserted in order: 5, 20, 30, 40, 50, 60, 70, 80, 90, 100, 160, 120, 15. (7)
- b) A binary tree has eight nodes. The post order and in order traversals of the tree are given below. Draw the tree. (8)
Post order: UTRWVSQP
In order: URT PQWSV

OR

Q7.

- a) Explain the insertion and deletion process of a node in Binary Search Tree with appropriate examples. (5)
- b) Explain the different traversal methods of a binary tree with appropriate examples. (5)
- c) The task is to find sum of all elements smaller than and equal to Kth smallest element. Demonstrate the program to solve the above given task. (5)

Unit-IV

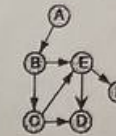
Q8.

- a) What do you mean by hashing? Discuss any two collision resolution techniques. (7)
- b) Create the adjacency list and adjacency matrix for the following directed graph: (8)

P.T.O.

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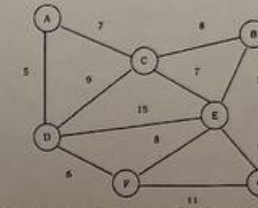
[-3-]



OR

Q9.

- a) Create a minimum spanning tree for the following graph using Kruskal's algorithm: (7)



- b) Differentiate between dense and sparse index with help of an example. (8)

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